

Evaluation of Generative Systems

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Introduction and Motivation

Classical Metrics

LLM-as-a-Judge

Multimodal Metrics

Human Evaluation

Benchmarks



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Benchmarks



- ▶ Generative AI is now widely deployed across domains (search, coding, healthcare, law, education, ...)
- ▶ Failures can have high real-world consequences
- ▶ Hallucinations → incorrect but confident outputs
- ▶ Bias → unfair or harmful outputs
- ▶ Evaluation determines trustworthiness
- ▶ Poor evaluation leads to unsafe deployment
- ▶ Good evaluation is needed for system tuning



- ▶ What does “correct” mean in generation?
 - Example: summarization → valid summaries
 - Example: chatbot → subjective quality
- ▶ Evaluation is ambiguous
- ▶ Ground truth may not exist
- ▶ Requires human judgment or proxies
- ▶ Non-deterministic output, sensitive to prompts
- ▶ Context-dependent behavior
- ▶ Multi-objective performance



- ▶ Small prompt changes → large output changes
- ▶ Prompt engineering affects evaluation results
- ▶ Benchmark prompts may bias outcomes
- ▶ Models may overfit prompts



- ▶ Many tasks have no unique answer
- ▶ Summaries differ in style and content
- ▶ Creative writing has infinite outputs
- ▶ Translation can vary
- ▶ → overlap-based metrics fail here
- ▶ → need for semantic evaluation
- ▶ → human judgment is often required



- ▶ Different annotators disagree
- ▶ Quality is subjective
- ▶ Cultural/contextual differences
- ▶ → requires annotation guidelines
- ▶ → inter-annotator agreement is needed
- ▶ Human annotations are expensive and slow but still a gold standard



- ▶ Correctness (refers to whether information is factually true, accurate, or on target)
- ▶ Relevance (refers to whether information is useful, on-topic, or significant to a specific need)
- ▶ Coherence (Logical Flow)
- ▶ Fluency (Smoothness)
- ▶ Safety
- ▶ Efficiency
- ▶ Robustness



► Intrinsic vs Extrinsic

- Intrinsic evaluation measures a system's internal components in isolation against predefined standards, while extrinsic evaluation assesses its performance based on real-world, downstream application success

► Offline vs Online

- Offline evaluation uses historical data to test models, online evaluation validates performance directly with live users

► Model vs System Evaluation

- Model evaluation focuses on assessing the raw performance of an AI model using metrics like accuracy, loss, or perplexity on a fixed dataset, whereas system evaluation assesses the entire application—including prompts, retrieval, tool calls, and user experience



- ▶ Static vs Dynamic
 - Static evaluation tests focuses on structure, code, or pre-defined datasets, dynamic evaluation tests the AI in real-time, observing its behavior during active execution or interaction
- ▶ Human vs Automated
- ▶ Task-Specific Evaluation
- ▶ Multimodal Evaluation



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		Predicted Class	
		P	N
Actual Class	P	True Positives (TP)	False Negatives (FN)
	N	False Positives (FP)	True Negatives (TN)

¹https://www.researchgate.net/figure/Structure-of-the-confusion-matrix-with-TP-FN-FP-and-TN-values_fig5_318859551



- ▶ **Accuracy:** The proportion of **true results** (both true positives and true negatives) among the total number of cases
- ▶ **Precision:** The fraction of **relevant instances** among the retrieved instances
- ▶ Precision measures how well retrieved documents match the specific information needed in a query
- ▶ In generative AI, high precision indicates the model produced sequences that are factually consistent with the target ground truth

Mathematical Formula for Precision

$$\text{Precision} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Positives (FP)}}$$

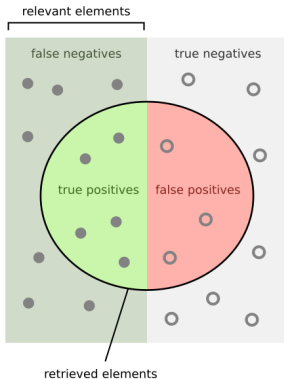


- ▶ **Recall:** The fraction of **relevant instances** that have been retrieved over the total amount of relevant cases
- ▶ Recall emphasizes how much of the reference information the system managed to capture

Recall Formula

$$\text{Recall} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Negatives (FN)}}$$

Precision vs. Recall



How many retrieved items are relevant?

$$\text{Precision} = \frac{\text{true positives}}{\text{true positives} + \text{false positives}}$$

How many relevant items are retrieved?

$$\text{Recall} = \frac{\text{true positives}}{\text{true positives} + \text{false negatives}}$$



- ▶ **F-measure (F1 Score)**: A traditional metric that emphasizes the significance of both **precision and recall** in determining response quality
- ▶ The F1 score is the default accuracy metric for tasks requiring **partial string overlap**, such as question answering or methods for checking number and precision of retrieved items

F1 Score (Harmonic Mean)

$$F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$



- ▶ **BLEU** is a traditional automated metric for assessing **text generation quality**
- ▶ It was originally developed to evaluate **machine translation** against human reference translations
- ▶ It measures the **content overlap** between a system's output and human-generated ground truths



- ▶ **N-gram Basis:** The metric calculates precision by matching sequences of n words (n-grams) between texts
- ▶ BLEU relies on **exact string matching** rather than embeddings

BLEU Score

$$BLEUScore = BP * \exp\left(\sum_{i=1}^N (w_i * \ln(p_i))\right)$$

- ▶ Where:
 - BP stands for Brevity Penalty
 - w_i is the weight for n-gram precision of order i (typically weights are equal for all i)
 - p_i is the n-gram modified precision score of order i .
 - N is the maximum n-gram order to consider (usually up to 4)



- ▶ BLEU applies a **brevity penalty** to discourage models from generating overly short responses
- ▶ Without this penalty, a model could achieve perfect precision by outputting just **one correct word**
- ▶ The penalty ensures the generated output is **comparable in length** to the human reference

$$BP = \exp\left(1 - \frac{r}{c}\right)$$

- ▶ Where:
 - r is the length of the candidate translation
 - c is the average length of the reference translations.



<https://www.geeksforgeeks.org/nlp/nlp-bleu-score-for-evaluating-neural-machine-translation-python/>



- ▶ **ROUGE** (Recall-Oriented Understudy for Gisting Evaluation) is an automated suite used to evaluate **text summarization quality**
- ▶ It assesses quality by comparing a system-generated output against human-authored **reference summaries**
- ▶ The metric serves as an indicator of the **token-level overlap** between the generated text and the ground truth
- ▶ While BLEU focuses on precision, ROUGE emphasizes how much of the **reference information** the system managed to capture



- ▶ **ROUGE-N**: Measures the overlap of **n-grams** (sequences of n words), such as unigrams or bigrams
- ▶ **ROUGE-L**: Evaluates similarity based on the **Longest Common Subsequence** (LCS)
- ▶ **ROUGE-S**: Focuses on **skip-bigram** overlap, measuring pairs of words regardless of their position in the sequence
- ▶ <https://www.geeksforgeeks.org/nlp/understanding-bleu-and-rouge-score-for-nlp-evaluation/>



- ▶ **METEOR** (Metric for Evaluation of Translation with Explicit ORdering) is a traditional automated metric for assessing the quality of machine-translated or generated text.
- ▶ Unlike metrics that rely solely on exact string overlap, METEOR accounts for **stems and synonyms** (using WordNet) to recognize semantic equivalence.
- ▶ It is based on the **harmonic mean of unigram precision and recall**, typically placing a higher mathematical weight on recall to ensure content coverage.
- ▶ It includes a **penalty for fragmentation** (the "chunking penalty") to reward generated sequences that maintain the same word order as the reference.
- ▶ METEOR was designed to achieve a **higher correlation with human judgment** than BLEU, particularly by addressing the limitations of exact n-gram matching.



- ▶ Perplexity is a metric that measures the uncertainty of a model's predictions.
- ▶ Specifically, in language models, it quantifies how well the model predicts the next word in a sequence.
- ▶ When a model makes a prediction, it assigns probabilities to possible next words.
- ▶ Entropy measures the level of uncertainty in the model's output.
- ▶ Lower entropy means the model is more certain about its predictions and therefore, the perplexity is lower.
- ▶ Perplexity indicates the level of confidence the model has in its prediction—lower perplexity suggests higher confidence and better performance in predicting the next word, while higher perplexity signals more uncertainty and less reliability.
- ▶ Perplexity represents the number of potential options the model is considering when making its prediction.



- ▶ **Prediction Accuracy:** Perplexity gives insight into the accuracy of a model's predictions. A low perplexity means the model is good at predicting words and likely generates coherent and fluent text.
- ▶ **Confidence of the Model:** It tells us how confident the model is in its predictions. If the perplexity is high, the model is likely uncertain about the next word, which could lead to incoherent text.

How is Perplexity Calculated?



- ▶ First, we need to compute the log probability of the model's predictions for each word in the sequence.
- ▶ Process:
 - Prediction of the Next Token: Language model predicts the probability of the next word based on the input text.
 - Logarithmic Transformation: Log of the probability is taken
 - Average Log-Likelihood: Average log-likelihood of all predicted words in the test set is computed.
 - Exponentiation to Get Perplexity: Final step is to exponentiate the average log-likelihood to get the perplexity score.



- ▶ Perplexity for a sequence of words can be computed as:

$$\text{Perplexity} = \exp \left(-\frac{1}{N} \sum_{i=1}^N \log p(w_i | w_{i-1}, w_{i-2}, \dots, w_1) \right)$$

- ▶ Where:
 - $p(w_i | w_{i-1}, \dots, w_1)$ is the predicted probability of the i^{th} word.
 - N is the total number of words in the sequence.
- ▶ This formula tells us how many words, on average, the model is choosing from when predicting the next word.
- ▶ A lower perplexity indicates fewer choices, meaning the model is more confident



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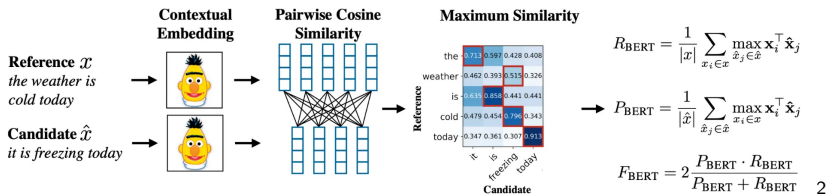


- ▶ **BERTScore** is an automated metric used for evaluating the quality of text generated by AI models
- ▶ It leverages **contextual embeddings** from pre-trained transformers, such as BERT, to evaluate text
- ▶ Unlike traditional metrics, it focuses on the **semantic similarity** between a system output and a reference text
- ▶ It was developed to address the "semantic gap" found in exact-match metrics like BLEU and ROUGE

How BERTScore Functions



- ▶ The metric computes similarity at the **token level** using embeddings
- ▶ The metric is highly sensitive to **semantic equivalence**, even when the model uses entirely different wording



²<https://medium.com/data-science/bertscore-evaluating-text-generation-with-bert-beb7b3431300>



- ▶ Unlike n-gram metrics, BERTScore requires a **forward pass** through a transformer, increasing latency
- ▶ Its accuracy is contingent on the **quality and domain** of the underlying pre-trained transformer model
- ▶ It remains a "black-box" compared to simple word counting
- ▶ Automated evaluations can sometimes fail to detect **subtle qualitative distinctions** that a human expert would notice

- ▶ COMET (Crosslingual Optimized Metric for Evaluation of Translation) is a learned evaluation metric trained to predict human judgments
- ▶ COMET learns a function: $f(\text{source}, \text{candidate}, \text{reference}) \rightarrow$ human-like quality judgment

- ▶ 1. Input Representation: Triplet (Source sentence s , Candidate output y , Reference r) are concatenated or jointly encoded
- ▶ 2. Encoder is typically based on multilingual models (e.g., XLM-R) and it produces contextual embeddings (h_s , h_y , h_r)
- ▶ 3. Feedforward network predicts $\hat{y}$ which is compared with y . Supervised learning is used to minimize error vs human scores
- ▶ The model captures Adequacy (meaning preservation), Fluency (language quality), and Alignment with source

What is LLM-as-a-Judge?



- ▶ A strong LLM is used to compare the outputs with ground truth, assign scores to the outputs and generate critiques
- ▶ Inputs typically include:
 - Prompt/task description
 - One or more candidate outputs
 - Evaluation rubric or criteria
- ▶ Output formats:
 - Scores (e.g., 1–10)
 - Pairwise preference (A vs B)
 - Structured feedback



- ▶ Phase 1: Surface metrics
 - BLEU, ROUGE
 - Assumption: similarity = quality
 - Limitation: ignores meaning, reasoning
- ▶ Phase 2: Learned metrics
 - BERTScore, COMET
 - Capture semantic similarity
 - Still reference-dependent
- ▶ Phase 3: LLM evaluators
 - No reference needed
 - Can evaluate: Coherence, Factuality, Helpfulness



- ▶ TruthfulQA: Uses models to classify truthfulness
- ▶ SelfCheckGPT: Detect hallucinations via consistency
- ▶ GPT-based evaluators (early GPT-3 work): Prompted to rate or classify outputs
- ▶ LLMs contain latent evaluative capabilities
- ▶ Limitation of early work:
 - Weak prompting
 - No structured rubrics
 - Low reliability

- ▶ Use chain-of-thought reasoning for evaluation
- ▶ Prompt structure:
 - Define criteria (e.g., coherence, fluency)
 - Ask model to reason step-by-step
 - Produce a final score
- ▶ Why it works:
 - Forces decomposition of judgment
 - Reduces shallow heuristics
- ▶ <https://deepeval.com/docs/metrics-llm-evals>



- ▶ 1. Pointwise scoring
 - Assign numeric value
 - Problems: Calibration issues, Scale interpretation varies
- ▶ 2. Pairwise comparison
 - Choose better of A vs B
 - More robust: Humans also prefer this format
- ▶ 3. Ranking
 - Sort multiple outputs
 - Computationally expensive
- ▶ 4. Critique-based
 - Generate explanations or feedback



- ▶ Fine-tuned judge models
 - Smaller models trained as evaluators
 - Trade-off: Efficiency vs generalization
- ▶ Fairness-aware judges
 - Reduce bias (e.g., position, verbosity)
- ▶ Consistency benchmarks (e.g., SAGE)
 - Evaluate: Stability, Self-consistency
- ▶ Test-time scaling (e.g., JETTS)
 - Compare: Judges vs reward models
- ▶ Multi-agent judging: Multiple LLMs voting or debating
- ▶ Process-based evaluation: Evaluate reasoning steps, not just outputs



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- ▶ **Retrieval Phase:** Precision, Recall, F-1, MAP, MRR
- ▶ **Generation Phase:** BLEU is often used to assess the generated answer
- ▶ **Faithfulness:** It evaluates if the response **accurately reflects** information retrieved from external sources
- ▶ **Correctness:** The metric compares responses to sample ground-truth answers to measure factual accuracy



- ▶ Core question: Is the output grounded in the input modality?
- ▶ Types of hallucination:
 - object hallucination (non-existent objects)
 - attribute hallucination
 - relational hallucination
- ▶ Metrics / datasets:
 - CHAIR (caption hallucination)
 - POPE (object hallucination detection)



- ▶ CLIPScore: measures image-text alignment
- ▶ FID (Fréchet Inception Distance): compares distribution of generated vs real images
- ▶ Human evaluation:
 - realism
 - prompt alignment
 - aesthetics



- ▶ Answers: “Does the image match the prompt?”
- ▶ Uses CLIP and encodes image into vector embedding and computes cosine similarity between embeddings
- ▶ Higher score → stronger semantic alignment
- ▶ Does not require reference captions
- ▶ Works for open-ended generation (e.g., text-to-image)



- ▶ Measures a distance between distributions of real and generated images
- ▶ Extract features using Inception v3
- ▶ Model both sets as Gaussians and compute Fréchet distance (difference in mean + covariance)
- ▶ Interpretation: Lower FID $\rightarrow$ generated images are closer to real distribution
- ▶ Strengths: Captures global image quality
- ▶ Limitations:
 - Ignores prompt/text alignment
 - Sensitive to dataset choice
 - Can be gamed



- ▶ Use multimodal LLMs (e.g., GPT-4V-style)
- ▶ Evaluate: grounding, factual correctness, explanation quality, ...
- ▶ Advantages: scalable, flexible
- ▶ Limitations: hallucinated evaluation, bias toward longer answers



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- ▶ Humans serve as the gold standard for common sense tasks
- ▶ Humans are essential for evaluating tasks where traditional ground truths are elusive or automated metrics fail to discriminate quality
- ▶ But there is no objective ground truth for many tasks
- ▶ Human reviewers exhibit significantly higher variability and standard deviation in their quality ratings compared to models



- ▶ Position bias (first answer preferred)
- ▶ Verbosity bias (longer = better)
- ▶ Cultural bias
- ▶ Anchoring bias
- ▶ Preference for fluency



- ▶ **Anthropomorphization:** Humans may inappropriately assign human-like feelings or motifs to AI interfaces, requiring careful tracking
- ▶ **Automation Bias:** Over-reliance can occur, where humans unjustifiably perceive AI content to be of higher quality than other sources
- ▶ **Algorithmic Aversion:** Conversely, a lack of system knowledge can lead to unnecessary aversion, depriving users of beneficial AI uses



- ▶ Crowdsourcing (cheap, noisy)
- ▶ Experts (expensive, high-quality)
- ▶ Synthetic users (LLMs)
- ▶ Trade-off: cost vs quality vs speed



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Benchmarks

What is a Benchmark?



- ▶ Dataset + task + metric
- ▶ Enables model comparison
- ▶ Drives research progress
- ▶ Classical Benchmarks:
 - GLUE / SuperGLUE (NLP): Designed to foster robust NLU models across nine tasks including single-sentence classification and similarity. SuperGLUE introduced to address GLUE limitations by providing more challenging reasoning tasks.
 - ImageNet (vision)
 - SQuAD (QA)



- ▶ MMLU: Tests knowledge and problem-solving across 57 academic and professional domains like law and math
- ▶ BIG-bench
- ▶ HELM: holistic evaluation



- ▶ GSM8K: Evaluates multi-step mathematical reasoning through grade-school word problems
- ▶ MATH: Contains 12,500 problems from high school competitions requiring step-by-step solutions
- ▶ ARC-Challenge: Assesses scientific reasoning through questions sourced from US grade-school science exams



▶ MT-Bench

- Multi-turn conversational benchmark
- <https://huggingface.co/spaces/lmsys/mt-bench>
- Evaluates: Coherence, Consistency across turns

▶ Chatbot Arena

- Crowdsourced pairwise comparisons
- <https://huggingface.co/spaces/lmarena-ai/chatbot-arena>
- <https://openlm.ai/chatbot-arena/>

▶ GPT-4-level judges achieve high agreement with humans



- ▶ VQAv2: large-scale visual QA dataset
- ▶ GQA: compositional reasoning
- ▶ TextCaps: captioning with OCR
- ▶ ScienceQA: multimodal reasoning (diagrams + text)
- ▶ MMMU: expert-level multimodal reasoning (math, medicine, etc.)
- ▶ SEED-Bench: diverse multimodal capabilities



- ▶ TruthfulQA: hallucination detection
- ▶ AdvBench: adversarial prompts
- ▶ MLCommons Safety Benchmark: Science-specific benchmarks for areas like cloud masking and materials material science



- ▶ Benchmarks $\neq$ real-world usage
- ▶ Training data leakage possible
- ▶ Real users ask harder questions
- ▶ Edge cases often ignored
- ▶ Evaluation datasets are stale, need for dynamic evaluation
- ▶ Many benchmarks are solved